On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

Kağan Argon
15071954

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① $A = \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$

a) Find the unknown entries α, γ, β, k and $m = ?$

$R_1 \cdot -3\alpha + R_2 \to R_2$
$-2R_1 + R_3 \to R_3$
$-R_1 + R_4 \to R_4$

$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 0 & 5-3\alpha^2 & -9\gamma+\gamma 3\alpha & 1-3\alpha^2 \\ 0 & \beta-2\alpha & 3+2\gamma & \alpha^2-2\alpha \\ 0 & 14-\alpha & -6+\gamma & 2-\alpha \end{bmatrix}$$

② $x=5y$ k dimensional

$x-5y=0$ $\begin{bmatrix}1 & -5 & 0\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}$

a) minimum dimension of the space is 3 dimensional because the intersection with another geometry gives volume so it should be 3 dimensional

b) $2x+y-z=4$

c) $x+4y-z=6$ Find all vectors

$\begin{bmatrix}1 & 4 & -1\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}6\end{bmatrix}$

$z=-6+4y+x$

$z=0$ $4y=6-x$

$y=1$ $x=2$

$y=0$

$z=x-6$

$z=2$ $x=8$

$N(A)=\begin{bmatrix}2\\1\\0\end{bmatrix}\cdot\alpha+\begin{bmatrix}8\\0\\2\end{bmatrix}\cdot\beta$

basis

(3) u is a unit vector $H = I - 2uu^T$;

$|u| = 1$ u R^n'de bir birim vektör olsun

$H_{n\times n}$ symmetric matrix

$H = I - 2uu^T$

a) $H^2 = (I - 2uu^T)(I - 2uu^T)$

$H^2 = I^2 - I \cdot 2u \cdot u^T - 2uu^T \cdot I + 4u(u \cdot u^T) \cdot u^T$

$= I(I - 2uu^T - 2uu^T) + 4u \cdot |u|^2 \cdot u^T$

$= I - 2uu^T - 2uu^T + 4u \cdot 1^2 \cdot u^T$

$H^2 = I - 4uu^T + 4uu^T$

$H^2 = I - 4 \cdot 1 + 4 \cdot 1 = I$

$\boxed{H^2 = I}$

$|u|^2 = u^T \cdot u$

$|u| = 1$

$|u| = \sqrt{u^T \cdot u}$

b) H is a symmetric matrix $\longrightarrow H = H^T$

$H \cdot H = H \cdot H^T$

$H^2 = H \cdot H^T$

$H \cdot H^T = I$

$\boxed{H \cdot H^T = H^T \cdot H = I}$

it is an orthonormal matrix

c) $(A - I \cdot \lambda) \cdot x = 0$

$$Hu = (I - 2u \cdot u^T) \cdot u = I \cdot u - 2u \cdot \underbrace{u^T \cdot u}_{|u|^2}$$

$$= I \cdot u - 2u \cdot |u|^2$$

↓

it is a identity matrix, yani birim matrisin tek eigenvalue değeri 1'dir

$I \cdot u = 1 \cdot u = u$ $\quad |u| = \sqrt{u^T \cdot u}$ $\quad u \rightarrow$ birim vektör

$u^T \cdot u = |u|^2 = 1$

$H \cdot u = u - 2u \cdot |u|^2 = u - 2u \cdot 1 = -u$

$Hu = \underbrace{(-1)}_{\downarrow} \cdot u$

λ eigenvalue is -1 $\boxed{\lambda = -1}$

d) $H = (I - 2u \cdot u^T)$

$H^T = (I - 2u \cdot u^T)^T$

$= I^T - 2(u \cdot u^T)^T = I - 2\, u^T \cdot (u^T)^T$ $\quad \boxed{(x \cdot y)^T = x^T \cdot y^T}$ $\quad \boxed{(x - y)^T = x^T - y^T}$

$\boxed{I^T = I}$

$= I - 2u^T \cdot u = H$

↳ H is a symmetric matrix

$H = S \Lambda S^{-1}$

$= (S^{-1})^{-1} \Lambda S^{-1}$

$= S^{-1} \Lambda S$

H is a diagonalizable matrix

④ $V(0)=10$ lt liquid $\quad w(0)=40$ Lt water

x 4x

30 $2x+x$ $3x$ $2x$ 20

25 25

a)

$$V_{k+1} = 2V_k - 2w_k$$
$$w_{k+1} = -V_k + w_k$$

İlk boşta ortadaki fark $30=3x$ diyelim

bunun $2x=20$'si birinci tanka geçer kalan x'i ikinci tanka

$30=3x$ $\quad$ 10 $3x$ 40

30 20

25 25

$$\frac{d}{dt}\begin{bmatrix} V \\ w \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} V \\ w \end{bmatrix}$$

b) $|A-\lambda I|=0$

$$\begin{vmatrix} 2-\lambda & -2 \\ -1 & 1-\lambda \end{vmatrix}=0$$

$2-2\lambda-\lambda+\lambda^2-2=0$

$\lambda^2-3\lambda=0 \quad \lambda_1=0 \quad \lambda_2=3$

$\lambda(\lambda-3)=0$

$\lambda_1 \rightarrow (A-0.I).x_1=0$

$$\begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix}.x_1=0 \qquad x_1=\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$\lambda_2 \rightarrow (A-3I).x_2=0$

$$\begin{bmatrix} -1 & -2 \\ -1 & -2 \end{bmatrix}.x_2=0 \qquad x_2=\begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} a_n \\ b_n \end{bmatrix} = c_1 x_1 . e^{\lambda_1 . t} + c_2 x_2 e^{\lambda_2 . t}$$

for $t=0$ $\quad c_1.\begin{bmatrix}1\\1\end{bmatrix}.e^0 + c_2.\begin{bmatrix}2\\-1\end{bmatrix}.e^{3.0} = \begin{bmatrix}10\\40\end{bmatrix}$

$$c_1 + 2c_2 = 10$$
$$-/c_1 + (-c_2) = 40$$

$3c_2=-30 \quad c_2=-10$

$c_1=30$

for $t=1$ $\quad u(k) = 30 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} \underbrace{e^{0.1}}_{1} + (-10)\begin{bmatrix} 2 \\ -1 \end{bmatrix} \cdot e^{3 \cdot 1}$

$e^3 \cong 20{,}08$

$$= 30 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + (-10) \begin{bmatrix} 2 \\ -1 \end{bmatrix} \cdot (20{,}08)$$

$$\begin{bmatrix} 30 - 10 \cdot 2 \cdot (20{,}08) \\ 30 + 10 \cdot 20{,}08 \end{bmatrix}$$

c) $t \to \infty \quad u(k) = 30 \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{0 \cdot \infty}}_{\text{steady state}} + (-10)\begin{bmatrix} 2 \\ -1 \end{bmatrix} e^{3 \cdot \infty}$